

Md Toufikul Islam Chowdhury

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EDUCATION

University of Greenwich	2025 M.Sc. , Data Science, with Merit Dissertation: 80% (distinction level) Modules include Statistical Methods for Time Series Analysis; Machine Learning; Applied Machine Learning; Big Data
Leading University	2022 B.Sc. , Computer Science and Engineering CGPA 3.30/4.00; 3.61/4.00 in the final two years Modules include Probability & Statistics; Artificial Intelligence; Neural Network & Fuzzy Logic; Discrete Mathematics; Differential Equations & Fourier Analysis; Numerical Methods

RESEARCH EXPERIENCE

My research combines geospatial accessibility modelling, spatial statistics, statistical modelling and machine learning to study access to health services and the health workforce in England, using national data from NHS England, the Care Quality Commission and the Office for National Statistics.

Independent Research	2026 – Present	Research Collaborator Maternity care accessibility With: Dr Jia Wang (University of Greenwich)
University of Greenwich School of Computing and Mathematical Sciences	2024 – 2025	M.Sc. Dissertation Research Workforce trend analysis, geovisualisation and SARIMAX forecasting for midwives and ambulance staff Supervisor: Dr Jia Wang

PAPERS AND MANUSCRIPTS

- Quality-adjusted spatial accessibility to maternity care in England: A provider-determined measure and the limits of neighbourhood-scale inference. Md Toufikul Islam Chowdhury and Jia Wang. In preparation for *Urban Science* (MDPI).
- Spatiotemporal workforce dynamics and forecasting in NHS England: Spotlight on midwives and ambulance staff. Md Toufikul Islam Chowdhury. M.Sc. dissertation, University of Greenwich, 2024–25.

COMPETITIONS

- CUSP London Data Dive, King's College London, urban-waste challenge (80+ students from eight universities), 2025.

MISCELLANY

- Methods: network-based nearest-provider accessibility; global and local bivariate Moran's I with catchment-restricted permutation inference and false discovery rate control; effective sample size from intraclass correlation; weighted least squares with cluster-robust standard errors; gradient boosting with spatially blocked cross-validation; SARIMAX time-series forecasting.
- Software: Python (pandas, NumPy, SciPy, statsmodels, scikit-learn, GeoPandas, PySAL, matplotlib, Plotly, Dash); OSRM; QGIS and PyQGIS; SQL; Git; $\text{\LaTeX}$.
- Data: Care Quality Commission ratings; ONS geographies, births and Census 2021; English Indices of Deprivation 2025; NHS England Maternity Services Data Set and Workforce Statistics.

REFERENCES

- Dr Jia Wang, Senior Lecturer in Spatial Data Science, School of Computing and Mathematical Sciences, University of Greenwich. j.wang@greenwich.ac.uk